Plagiarism Scan Report

5% Plagiarism

5% Exact Match

0% Partial Match

95% Unique

Words	1000
Characters	6125
Sentences	33
Paragraphs	141
Read Time	5 minute(s)
Speak Time	7 minute(s)

Content Checked For Plagiarism

MUNICIPAL WASTE COLLECTION
SCHEDULING & ROUTE
OPTIMIZATION SYSTEM
A Python-Based Prototype for Civic Waste-Management Operations
CSA08 · Programming in Python | SDG 11 — Sustainable Cities and Communities
P R E S E N TAT I O N R OA D M A P
Agenda
• 1. Problem Statement
• 2. Objective
• 3. Requirements & Environment
• 4. Design / Proposed Solution
• 5. Algorithm / Pseudocode / Flowchart
• 6. Implementation Overview
• 7. Test Cases & Expected Results
• 8. Sample Execution Output
• 9. Analysis & Discussion
• 10. Conclusion
• 11. Individual Contribution
• 12. References
Municipal Waste Collection Scheduling and Route Optimization System 2
0 1 · P R O B L E M S TAT E M E N T
Why Manual Waste-Collection Management Fails
• Data inconsistency — zone names, truck IDs and timings recorded differently across ward offices
• Inefficient search — locating a truck's zone or a zone's last collection date takes significant time
• Missed pickups — zones skipped with no systematic tracking, causing garbage pile-up and complaints
• Unbalanced truck load — some trucks serve far more stops than others, causing delays and overtime
• No performance analysis — high-missed-pickup zones, truck workload and complaint trends stay invisible
• Manual, redundant, error-prone reporting and scheduling across paper logs and registers
Municipal Waste Collection Scheduling and Route Optimization System 3
0 2 · O B J E C T I V E
Project Objective
Design, implement and evaluate a Python-based prototype that lets a municipal corporation manage waste-collection zones, truck schedules, and complaint/missed-pickup records efficiently — and generate meaningful civic performance reports.
Register
Zones and trucks with
structured, consistent

records
Schedule
Assign trucks to zones by
day/shift with load limits
Track
Log missed pickups and
complaints, then resolve
them
Analyze
Report workload,
coverage and complaint
trends

03 · REQUIREMENTS & ENVIRONMENT USED
Requirements & Development Environment
Functional Requirements
• Register zones (ID, area, ward, collection day) and trucks (ID,
capacity, driver)
• Assign trucks to zones for a given day/shift
• Log missed pickups and complaints; mark them resolved
• Search zones, trucks and complaints by name, ward or ID
• Generate reports: missed pickups, workload, resolution trend,
coverage
• Save/load data via CSV; handle invalid input without crashing
Environment & Tools
• Language: Python 3.x
• IDE: VS Code / PyCharm / IDLE
• Version control: Git & GitHub
• Core libraries: csv, datetime, statistics
• Testing: manual test cases / assertions
• Target scale: 300 zones, 100 truck schedules

04 · DESIGN / PROPOSED SOLUTION
Data Structure Design
Entity Structure Why
Zones Dictionary — zone_id : {area, ward, truck_id, last_collected} O(1) lookup by Zone ID
Truck Schedules Dictionary — truck_id : {route: [...], shift} Fast access to a truck's route &
shift
Complaints / Missed Pickups List of dicts — {zone_id, date, status} Preserves chronological order
Ward & Zone Type Tuple — (ward_no, zone_type) Fixed data, protected from
accidental change
Shift Types / Wards Set — {Morning, Afternoon, Evening} Guarantees uniqueness

04 · DESIGN / PROPOSED SOLUTION
System Architecture — Program Flow
Main
Menu → Zone & Truck
Registration → Schedule
Assignment → Collection
Logging → Complaint
Management → Analysis &
Reports → File
Operations
Data persists throughout via CSV files (zones.csv, truck_schedule.csv, complaints.csv), loaded at startup and

saved on exit or
on demand.

Municipal Waste Collection Scheduling and Route Optimization System 7

05 · ALGORITHM / PSEUDOCODE

Missed-Pickup Detection Algorithm

```
FUNCTION check_missed_pickups(scheduled_day):
  missed = []
  FOR zone_id, details IN zones.items():
    IF details['collection_day'] == scheduled_day:
      IF details['last_collected'] != today():
        missed.append(zone_id)
        complaints.append({
          'zone_id': zone_id, 'date': today(),
          'status': 'Missed Pickup'
        })
  RETURN missed
```

Scans every zone scheduled for the day and auto-logs a complaint if it was not collected.

Municipal Waste Collection Scheduling and Route Optimization System 8

05 · ALGORITHM / PSEUDOCODE

Truck Assignment & Load-Balancing Algorithm

```
FUNCTION assign_truck_to_zone(truck_id, zone_id, max_zones_per_truck=10):
  IF truck_id NOT IN truck_schedule:
    truck_schedule[truck_id] = {'route': [], 'shift': 'Morning'}
  current_load = LENGTH(truck_schedule[truck_id]['route'])
  IF current_load >= max_zones_per_truck:
    RAISE TruckOverloadError
  truck_schedule[truck_id]['route'].append(zone_id)
  zones[zone_id]['truck_id'] = truck_id
  RETURN True
```

Enforces a per-truck capacity cap and raises a custom exception instead of silently overloading a route.

Municipal Waste Collection Scheduling and Route Optimization System 9

06 · IMPLEMENTATION OVERVIEW

Modular Implementation Plan

zone_manager.py Zone CRUD operations
truck_manager.py Truck registration & schedule assignment
complaint_manager.py Complaint & missed-pickup logging
analytics.py Performance analysis functions
reports.py Report generation
file_handler.py CSV read / write operations
main.py Main menu & program flow

Municipal Waste Collection Scheduling and Route Optimization System 10

07 · TEST CASES & EXPECTED RESULTS

Minimum Test Scenarios

Scenario Expected Result
TC1 Register a new zone with valid details Zone added successfully
TC2 Assign a truck within its capacity limit Zone added to truck's route successfully
TC3 Assign a truck beyond max zone capacity TruckOverloadError raised & handled gracefully
TC4 Missed-pickup check for an uncollected zone Zone flagged; complaint auto-logged
TC5 Mark a logged complaint as resolved Status updated with resolution date
TC6 Search zones by ward number Matching zone records returned
TC7 Load a missing CSV file FileNotFoundError

Matched Source

Similarity 2%

Title:GitHub

SOFTWARE REQUIREMENTS. Language: Python 3.x; IDE: VS Code / PyCharm / IDLE (bundled with Python); Standard Library Modules: math , statistics; OS: Windows 10 ...

https://google.com/goto?url=CAESZAHrOzAVqBdXYpVcwPqm1tSYT0HSMf-RvqJZSrSPL0ug6LkAdUXITmaK-5GiOcvjJrK2xoMEkQm_j42sjzWEM3g-AmAYGotSgBuAnFeqUFuZ7P_9ibxAbxOwX2Q3qqfJsiCGIL0

Similarity 2%

Title:JavaScript syntax: what is ({ }) Is it a function or object?

as a developer with OOP background(c#, Java) OOP JavaScript is wild horse for me. I am trying learn nuts and bolts of language and then jump on libraries ...

Check

Similarity 2%

Title:Why aren't vehicles loading to maximum capacity in Truck ...

I have a fairly large long haul that I can't get my trucks to depart. I've even bought a new truck but it won't let me assign that truck to the ...Read more

Check